

Privacy by Choosing Less

Project: Helpful Classroom Sorter

1. Privacy by Choosing Less

Teach the learning cycle

Primary target: Justify which information a stated task needs and remove unnecessary fields

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Fair for Whom?

Without opening the old lesson, explain one key idea from Fair for Whom?. Give an example and a mistake that the idea helps you avoid.

Look for: Compute a correct-out-of-assessed fraction and explain its denominator

Reference example, not the only acceptable wording: A has 9 correct out of 12 assessed: $9/12 = 3/4 = 75\%$. The denominator includes all 12 judged outcomes, including 3 incorrect ones. Using $9/9$ would exclude the errors and falsely report perfect assessed accuracy.

Good Data, Bad Data

Without opening the old lesson, explain one key idea from Good Data, Bad Data. Give an example and a mistake that the idea helps you avoid.

Look for: Check records against a stated format, unit and meaning before using them

Reference example, not the only acceptable wording: Height 60 fails the whole-number range 1 through 10. Travel blank is missing; its value cannot be inferred. Travel 0 is valid in the whole-number range 0 through 10 and can represent observed no travel. Zero is a measurement, not a replacement for a blank.

Confidence Is Not Correctness

Without opening the old lesson, explain one key idea from Confidence Is Not Correctness. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish a confidence claim from evidence that a prediction was correct

Reference example, not the only acceptable wording: The learner replaced the documented score meaning with a probability interpretation. Report 80 of 100 feature-match points. To claim a chance of correctness, a defined probability method and relevant outcome evaluation would be needed.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A book recommender requests genre, maximum pages, home address and a selfie. Keep only fields needed for genre and length matching. Explain one consequence of storing extra fields.

Concept to explain

A useful system should collect only information it truly needs.

Term · Meaning

Term: privacy

Meaning: control and protection of information about people

Term: data minimization

Meaning: collecting only what the purpose requires

Term: purpose

Meaning: specific reason information is used

Term: retention

Meaning: how long information is kept

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach privacy as purposeful information design, not a demand that pupils disclose private experiences. Use fictional paper examples throughout; no account creation, real survey or upload is necessary. These examples are classroom design exercises, not legal advice or a substitute for school policy. Coded records with a re-identification key remain linkable. Aggregation reduces individual detail but does not guarantee anonymity in every context.

Readiness answer

No. It needs the assessed outcomes and counts, not a student's identity or home address. The required information depends on the purpose.

Fictional topic slips or counters marked drawing, music and coding

A sample form containing only the fictional field labels in W1

Paper for the five-question data plan; no real pupil or borrower records

End goal for the pictorial case

By the end, I can collect only information necessary for a stated task and explain coded-record limits.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Follow the information instead of imagining magic inside the machine.

INPUT candidate information fields

PROCESS keep only fields needed for the stated goal

OUTPUT useful result with reduced privacy risk

CORE IDEA A useful system should collect only information it truly needs.

UNDER THE HOOD Data minimization limits collection, access, retention, and reuse to what is necessary for a defined purpose. Less data reduces breach exposure and misuse, but essential high-stakes information still needs secure, consent-aware handling.

Draw a fresh example with all three stages:

Every step connects a claim to observable evidence.

CASE A lunch helper needs food preferences and allergy information for its task, not a face photo, home address, or family income.

STEP 1 — OBSERVE candidate information fields

STEP 2 — TRACE keep only fields needed for the stated goal

STEP 3 — CONCLUDE useful result with reduced privacy risk

MODEL REASONING Keep only information required for safety and stated preferences, allow a person to provide or update it, restrict access, and delete it when the purpose ends. Explain why each removed field adds risk without benefit.

DO NOT OVERCLAIM Useful does not mean permitted forever or for every new purpose.

What new evidence could change this answer?

Concrete teaching case

A genre-based book helper asks for genre, home address and a photograph.

Which fields are necessary?

Exact explanation to model

Genre is relevant; address and photo are unnecessary for the stated task. Collecting less reduces exposure.

Explain the added worked example

Privacy includes having appropriate boundaries around information about people. A useful first question is: what exact job are we trying to do? A second is: what is the least information that can do that job? Collecting extra details just because a form has room for them creates more information to protect.

Our fictional class wants to choose one of three workshop topics for Friday: drawing, music or coding. It needs a total for each topic, not a list of who chose what. Each fictional pupil gets one paper slip and marks one topic. No real class survey is needed for this lesson. The teacher counts the choices and uses the largest total; a tie goes to a class discussion.

A proposed form asks for name, full date of birth, home address, photograph and topic choice. For the stated counting task only the topic choice is needed. Remove the other fields before collection. Do not fill them in and promise to ignore them later. If the purpose changes to individual attendance, design and justify that separate task with the teacher rather than silently reusing the survey.

Suppose the fictional counts are drawing 8, music 7 and coding 5. These totals support the choice of drawing. There is no need to publish each slip or a photograph of a child holding it. The teacher checks the total of 20 against the number of slips and then follows the agreed school process for disposing of the slips after verification.

A retention plan needs a clear trigger and a responsible person: for example, the teacher disposes of the fictional vote slips after the totals are checked and the activity closes. Real classroom records must follow the school's approved process. Deleting one visible copy does not prove that every backup, screenshot or previously shared copy is gone.

Key terms

Term · Meaning

Term: Personal data

Meaning: Information about an identifiable person, including combinations that can identify someone

Term: Purpose

Meaning: The specific job information is collected or used to do

Term: Data minimisation

Meaning: Using only the information needed for a stated purpose

Term: Access control

Meaning: A rule and mechanism deciding who can see or use information

Term: Retention

Meaning: How long information is kept before an agreed deletion or other approved action

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Replace a name with P07 · Hides the name on this sheet · A separate key can reconnect the code

Replace a name with P07: Remove names but keep photos

Hides the name on this sheet: Removes one identifying field

A separate key can reconnect the code: Faces may still identify people

Replace a name with P07: Share topic totals only

Hides the name on this sheet: Removes individual rows

A separate key can reconnect the code: Small or unusual groups may still reveal a choice

A code is not a promise of anonymity. If a teacher keeps a key connecting P07 to a pupil, the coded record can still be linked to that pupil. Even without a key, a combination such as an exact birthday, a team and a photograph may identify someone. Ask whether a reader could connect the information to a person, not just whether the name column is blank.

Totals also need context. A report saying that the only member of a tiny club chose music reveals that member's choice. Our whole-class topic totals reveal less detail than individual rows, but we should not describe every possible total as automatically anonymous. Share only what the audience needs for the purpose.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Privacy by Choosing Less

A fictional teacher wants to count broken craft tools by type, without contacting borrowers.

The facts we will use

Needed output: totals for scissors, ruler and hole-punch repairs.

Draft form requests pupil name, birthday, tool type and repair-needed Yes/No. No borrower follow-up is required.

Privacy by Choosing Less

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. State purpose

Count repairs by tool type, not by person.



2. Choose fields

Retain tool type and repair-needed status.



3. Remove excess

Name and birthday are unnecessary for this stated task.



4. Report totals

Aggregate each tool type's Yes count; record the authorised retention plan.

CHANGE ONE THING

The teacher adds an authorised task to return one repaired tool to its borrower.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. State purpose

What happens and why: Count repairs by tool type, not by person.

Step: 2. Choose fields

What happens and why: Retain tool type and repair-needed status.

Step: 3. Remove excess

What happens and why: Name and birthday are unnecessary for this stated task.

Step: 4. Report totals

What happens and why: Aggregate each tool type's Yes count; record the authorised retention plan.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A field is justified by what the task needs to do, not by whether it might become useful someday. Names and birthdays do not help these repair totals. Replacing names with codes still leaves linked personal records if someone keeps a name-code key; collecting less can remove the need for that link entirely.

What this example does not establish

Removing these fields is not a universal guarantee of anonymity or permission to collect other identifying combinations.

Change one thing in the pictured case

Change: The teacher adds an authorised task to return one repaired tool to its borrower.

Prediction: A minimal contact mechanism may now be needed for that tool.

Reason: The purpose changed; reassess necessity rather than restoring every old field.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: A genre-based book helper asks for genre, home address and a photograph.

Write the question to explain: Which fields are necessary?

Reveal after predictions: Genre is relevant; address and photo are unnecessary for the stated task. Collecting less reduces exposure.

Keep for an independent attempt: A book helper needs reading topic and approximate difficulty. Someone asks it to collect home address too. Which information can be left out, and why?

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

A chart needs the three topic totals, not the named spreadsheet. Use the fictional counts provided here. Before any real tool use, check teacher approval for both tool and data; a smaller input does not itself authorize an upload. Draw bars on paper or use an approved local chart tool. Exit: collect topic choice; remove any two identity/contact fields. Share totals with the class. The teacher disposes of slips after verification and task closure under the agreed school process.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start 0 to 7 minutes

Teacher action: Retrieve which fields a performance count actually needs. Establish fictional examples only.

Stage: 7 to 20 minutes

Teacher action: Define the vote purpose. Complete W1 and remove unnecessary fields before collection.

Stage: 20 to 33 minutes

Teacher action: Trace collection, access, sharing and disposal on the diagram. Check the totals.

Stage: 33 to 45 minutes

Teacher action: Complete W2. Contrast coded records with a claim of anonymity.

Stage: Session 2 start 0 to 8 minutes

Teacher action: Retrieve why a blank name column does not prove anonymity.

Stage: 8 to 23 minutes

Teacher action: Complete W3 and the guided chart task. Discuss small-group inference and approved tool rules.

Stage: 23 to 38 minutes

Teacher action: Collect W4 independently. Check that purpose, fields, access, output and disposal align.

Stage: 38 to 45 minutes

Teacher action: Complete W5 and the exit. Rehearse telling a trusted adult without repeating exposed information.

Reduce help deliberately

Completion task: A teacher replaces names with codes and keeps a separate code-to-name key. A pupil calls the records anonymous. Explain the problem. Then explain why removing names while leaving photographs would not settle the question either.

Teacher check: The key can reconnect each code to a person, so the coded records are not anonymous just because names are absent. Restricting the key helps limit access but does not erase the link. Photographs can identify people even if the name field is removed.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Privacy and data minimisation. Proposed explanation: “More data always improves a task”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Unnecessary identity fields do not improve a category count and create more information to manage.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more

demanding variation.

Pictorial practice question

For the original task, does adding an anonymous-looking pupil code improve the repair count?

Reasoned teacher answer

No. The totals need tool type and repair status, not pupil identity. A linkable code can also retain personal-data risk.

Changed-case reasoning: A minimal contact mechanism may now be needed for that tool. The purpose changed; reassess necessity rather than restoring every old field.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Predict first, test honestly, and explain what the evidence means.

State your prediction before the result is known.

Complete the task: Redesign a lunch helper that asks for too much personal data.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your prediction.

Explain the result, one limitation, and the next useful test.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

CLAIM TO REPAIR Collecting extra personal details always improves a tool.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

More data always improves a task: Unnecessary identity fields do not improve a category count and create more information to manage.

Codes or missing names mean anonymous: A key or other identifying fields may reconnect records to a person.

Every total is safe to publish: A small or distinctive group may allow a reader to infer a person's choice.

Deleting one copy proves all copies are gone: Other copies, screenshots or backups may remain; follow the adult-led school process.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Genre is relevant; address and photo are unnecessary for the stated task. Collecting less reduces exposure.

Independent question

A book helper needs reading topic and approximate difficulty. Someone asks it to collect home address too. Which information can be left out, and why?

Expected independent answer

Leave out home address because it is unnecessary for the stated recommendation task. Collect only information that serves the learning purpose.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “Collecting extra personal details always improves a tool.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Keep genre and maximum pages. Address and selfie do not help the stated task and create unnecessary exposure if shared or lost. Use fictional records for practice and specify who may see needed data and

when it is deleted.

Explanations for the additional workbook examples

W1 Trim a form to its purpose

The fictional workshop vote asks for name, full date of birth, address, photograph and topic choice. Its only purpose is to count votes for drawing, music and coding. State which field to keep, which to remove and why. Explain why collecting the extra fields first is different from never collecting them.

Keep topic choice; remove name, full date of birth, address and photograph. The totals do not require identity or contact details. Collecting extra fields creates copies and exposure to manage even if those fields are later ignored or deleted.

W2 Challenge a claim of anonymity

A teacher replaces names with codes and keeps a separate code-to-name key. A pupil calls the records anonymous. Explain the problem. Then explain why removing names while leaving photographs would not settle the question either.

The key can reconnect each code to a person, so the coded records are not anonymous just because names are absent. Restricting the key helps limit access but does not erase the link. Photographs can identify people even if the name field is removed.

W3 Share the result the audience needs

The fictional class has drawing 8, music 7 and coding 5. Write the result needed to choose a workshop. A pupil suggests posting every vote slip online as proof. Explain a lower-exposure way to check and share the result. Why can a total about a one-person club still reveal a choice?

Drawing leads with 8; totals are 8, 7 and 5, adding to 20. The teacher checks the slips and totals, then shares the three totals with the class rather than public individual slips. A one-person club total can be linked to its only member, revealing that person's choice.

W4 Design a different minimal-data task

A fictional library wants only the total number of books needing repair in each category: cover, page or binding. It does not need to contact borrowers. A draft repair sheet asks for borrower name, home address, book repair category and borrower photograph. Design a plan: state the purpose, required and removed fields, who handles source sheets, what totals are shared with the librarian, and who disposes of sheets at what agreed point. Use no real borrower details.

Purpose: count repair needs by category. Keep repair category; remove borrower name, address and photograph because no borrower contact is required. An assigned teacher or librarian handles the source sheets, checks counts and shares category totals with the librarian. Name that adult as responsible for disposal after totals are verified and the counting task closes, following the agreed school process. Other justified restricted roles and clear disposal triggers are acceptable.

W5 Respond to an unnecessary upload

Someone plans to upload the library repair sheet with borrower details to an unapproved AI tool to add three totals. Explain a smaller-input alternative and a no-upload method. If the sheet was already shared, state a suitable first response and one claim you must not make about deleting it.

Use only fictional category counts in this lesson; for real work check teacher-approved tools and data rules before any upload. Add totals on paper or with an approved local calculator. If shared, stop further sharing and tell the teacher or trusted adult promptly so the school process can be followed. Do not claim that deleting the visible message removes all copies or backups.

Assess independent transfer

Selected case: A fictional library wants only the total number of books needing repair in each category: cover, page or binding. It does not need to contact borrowers. A draft repair sheet asks for borrower name, home address, book repair category and borrower photograph. Design a plan: state the purpose, required and removed fields, who handles source sheets, what totals are shared with the librarian, and who disposes of sheets at what agreed point. Use no real borrower details.

Selected case answer

Purpose: count repair needs by category. Keep repair category; remove borrower name, address and photograph because no borrower contact is required. An assigned teacher or librarian handles the source sheets, checks counts and shares category totals with the librarian. Name that adult as responsible for disposal after totals are verified and the counting task closes, following the agreed school process. Other justified restricted roles and clear disposal triggers are acceptable.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Helpful Classroom Sorter

Cross out names, faces, and ownership details unless an explicit need survives review; set a deletion rule for sorter records.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two mechanisms with the same visible output and explain what evidence distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 25: Generate, Retrieve, or Calculate?. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Offer a five-box planning frame: purpose, fields, access, shared result, disposal. Accept spoken explanations of why each field is or is not needed. Avoid asking children to describe past privacy incidents. Extension: construct a fictional small-group total that reveals one choice and suggest a less detailed output.

W4 requires a category-count purpose, repair category only from the listed fields, removal of all three borrower fields, a restricted responsible adult, category totals for the librarian and an actionable disposal trigger under the agreed process. After support change the task to contacting borrowers about repairs. Ask the pupil to explain that the old minimal field set no longer performs the new task and that an adult must design an approved contact workflow; do not request real contact details.

Week 25 distinguishes generating a plausible response, retrieving an existing record and calculating a result from explicit rules.

Plan the delayed return

Suggested return points in this level: ai-1/25, ai-1/27, ai-1/30. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Write the sorter's purpose, necessary fields, access rules and deletion point before collecting any real classroom information.

7. Prepare, teach and assess

Teaching target: Justify which information a stated task needs and remove unnecessary fields

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: A teacher replaces names with codes and keeps a separate code-to-name key. A pupil calls the records anonymous. Explain the problem. Then explain why removing names while leaving photographs would not settle the question either.

Reasoned practice answer: The key can reconnect each code to a person, so the coded records are not anonymous just because names are absent. Restricting the key helps limit access but does not erase the link. Photographs can identify people even if the name field is removed.

Check every learner before discussion: Use the worked case for Privacy and data minimisation. Proposed

explanation: “More data always improves a task”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Unnecessary identity fields do not improve a category count and create more information to manage.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The teacher adds an authorised task to return one repaired tool to its borrower. Require a prediction and a reason before comparing. Expected result: A minimal contact mechanism may now be needed for that tool. Reason: The purpose changed; reassess necessity rather than restoring every old field.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner’s explanation and evidence.

8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.